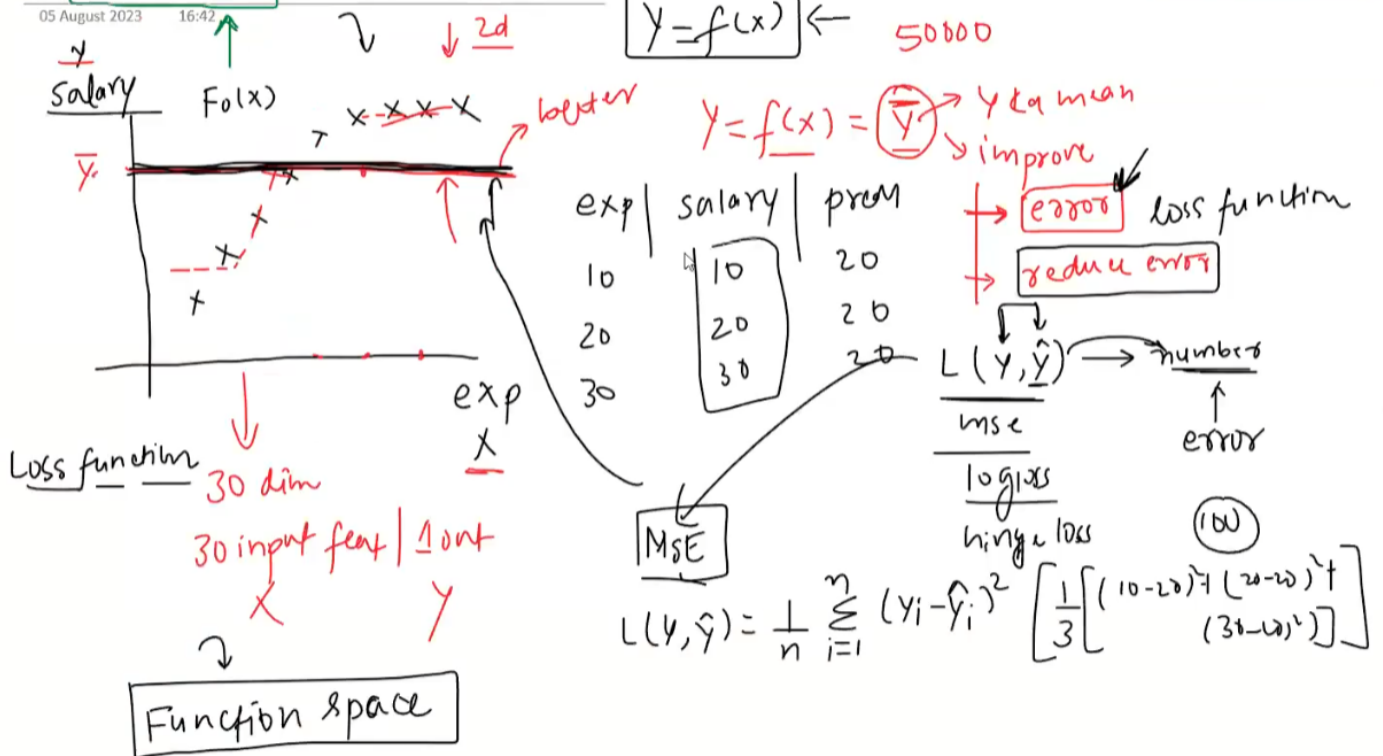


Gradient Boosting is performing Gradient Descent in Function Space

1. Function Space Vs Parameter Space

05 August 2023 16:42

Host



$$L(y, \hat{y}) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)$$

$$L(y, \hat{y}) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \underbrace{(y_1 - \hat{y}_1)^2}_{\text{variables}} + \underbrace{(y_2 - \hat{y}_2)^2}_{\text{variables}} + \underbrace{(y_3 - \hat{y}_3)^2}_{\text{variables}} + \dots + \underbrace{(y_n - \hat{y}_n)^2}_{\text{variables}}$$

Constants

Host

variables → may vary (model) type function

$$L(\hat{y}_1, \hat{y}_2, \hat{y}_3, \dots, \hat{y}_n)$$

math funct

how many dims

$n+1$ ✓

Loss function

$\hat{y}_i$ (n items)

$\hat{y}_1$
 $\hat{y}_2$

n+1 dimensional relationship

$$L(\hat{y}_1, \hat{y}_2, \hat{y}_3, \dots, \hat{y}_n)$$

how many dims

$$n+1 \checkmark$$

math funct

Loss function

$\hat{y}_i$ (n items)

n+1 dimensional relationship

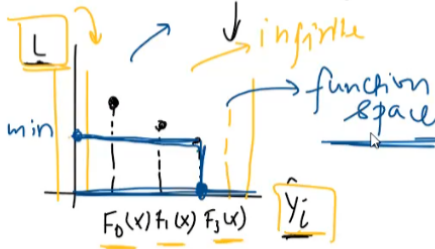
function space

L

$\hat{y}_1$

true function

n+1 dim graph



$$F_0(x) = \bar{y} = 20$$

$$\hat{y}_i = F_0(x) = \bar{y} = 20$$

1. Initialize $f_0(x) = \arg \min_{\gamma} \sum_{i=1}^N L(y_i, \gamma)$.

2. For $m = 1$ to M :

(a) For $i = 1, 2, \dots, N$ compute

$$r_{im} = - \left[\frac{\partial L(y_i, f(x_i))}{\partial f(x_i)} \right]_{f=f_{m-1}}.$$

(b) Fit a regression tree to the targets r_{im} giving terminal regions R_{jm} , $j = 1, 2, \dots, J_m$.

(c) For $j = 1, 2, \dots, J_m$ compute

$$\gamma_{jm} = \arg \min_{\gamma} \sum_{x_i \in R_{jm}} L(y_i, f_{m-1}(x_i) + \gamma).$$

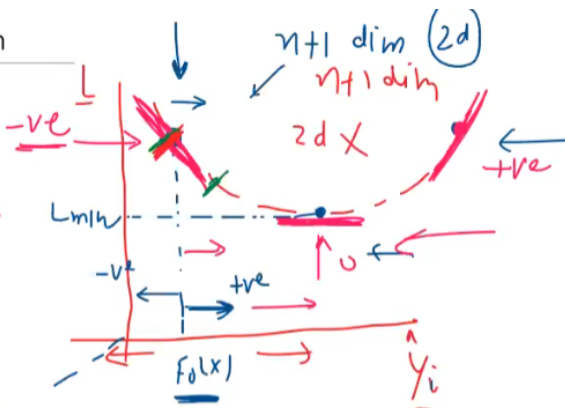
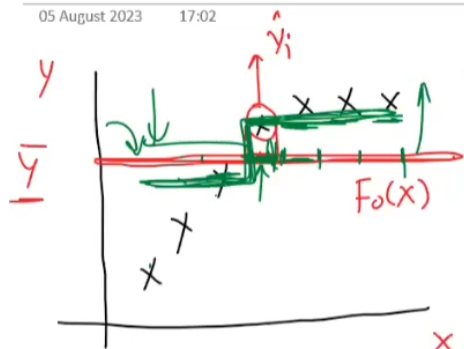
(d) Update $f_m(x) = f_{m-1}(x) + \sum_{j=1}^{J_m} \gamma_{jm} I(x \in R_{jm})$.

3. Output $\hat{f}(x) = f_M(x)$.

2. Direction of Loss Minimization

05 August 2023 17:02

Host



slope +ve
slope -ve

slope

- slope

$$L = (y_i - \hat{y}_i)^2$$

$$\left[\frac{\partial L}{\partial \hat{y}_i} \right]$$

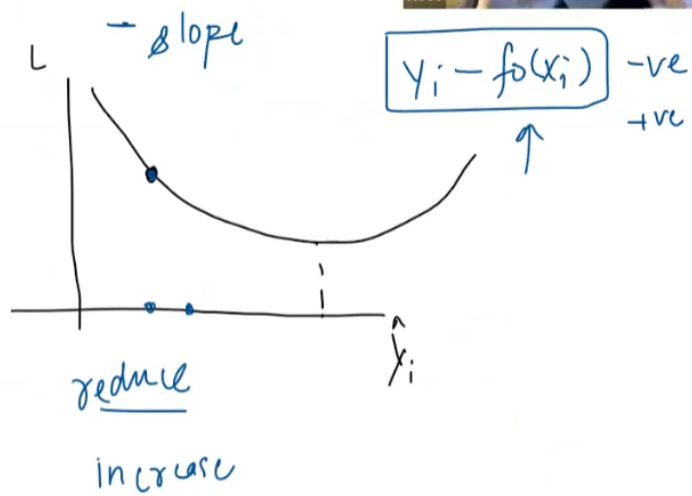
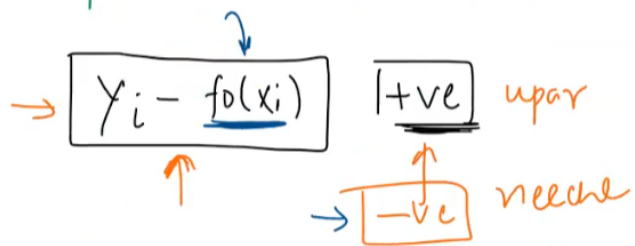
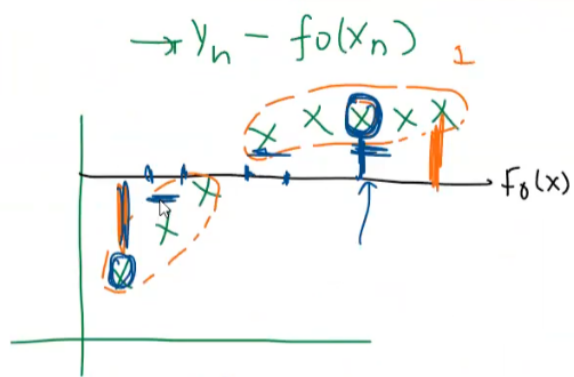
- slope

n slope n slope n number

$$\hat{y}_i = F_0(x_i)$$

$$\frac{\partial L}{\partial F_0(x_i)}$$

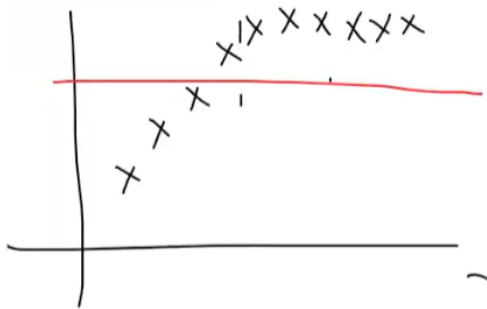
$$\left[\frac{\partial L}{\partial \hat{y}_1} \quad \frac{\partial L}{\partial \hat{y}_2} \quad \frac{\partial L}{\partial \hat{y}_3} \quad \dots \quad \frac{\partial L}{\partial \hat{y}_n} \right]$$



3. Update the Function

05 August 2023 17:15

Host



$F_0(x)$
update



$$\left[\frac{\partial L}{\partial \hat{y}_i} \right]$$

direction

$$y_i - f_0(x_i)$$

$$m_{\text{new}} = m_{\text{old}} -$$

slope

$$F_1(x_i) = F_0(x_i) - \text{slope}$$

$$F_1(x_i) = F_0(x_i) + y_i - f_0(x_i)$$

$$F_0(x) = m_{\text{old}}$$

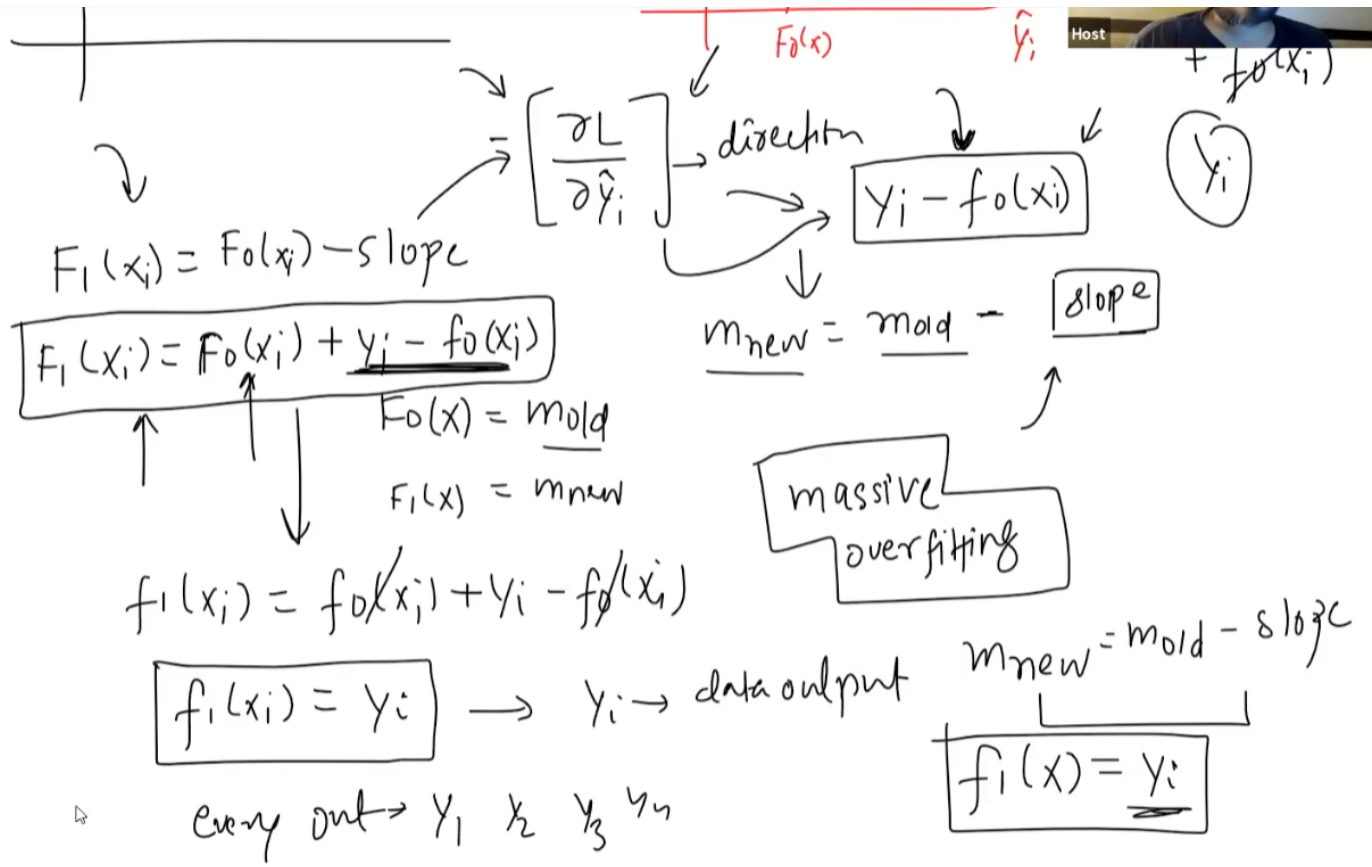
$$F_1(x) = m_{\text{new}}$$

$$f_1(x_i) = f_0(x_i) + y_i - f_0(x_i)$$

massive
overfitting

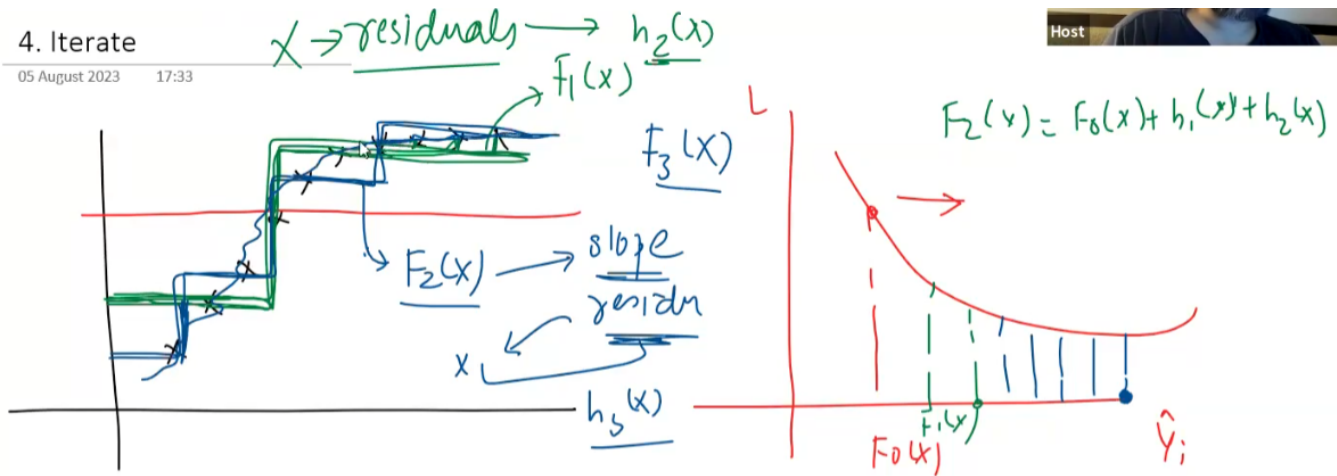
Host

$+ f_0(x_i)$



4. Iterate

05 August 2023 17:33



$$F_1(x) = F_0(x) + \underbrace{h_1(x)}_{\text{weak den}}$$

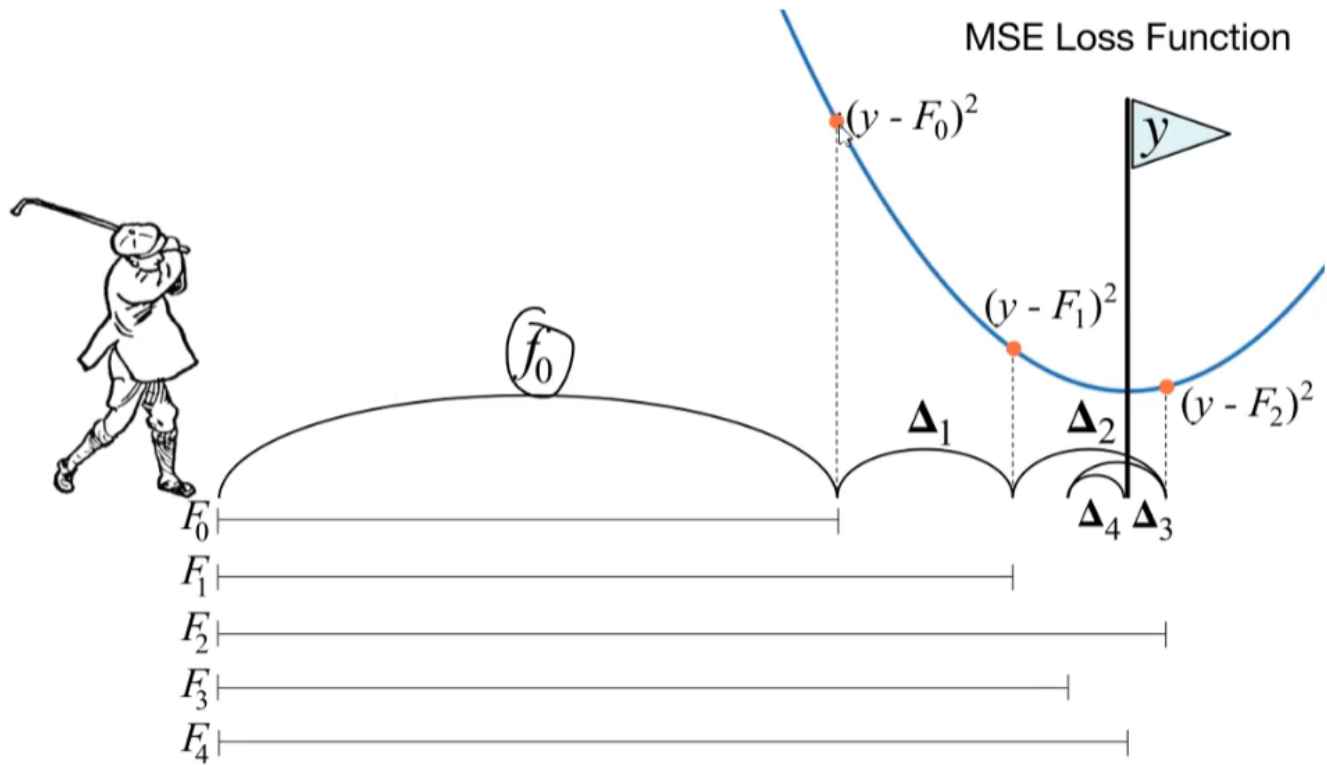
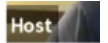
$x | \text{residual}$



Another Perspective

05 August 2023

07:58



Difference Between Gradient Boosting and Gradient Descent

Host

04 August 2023 15:22

[1] Gradient Descent $\rightarrow$ parameter space
Gradient Boosting $\rightarrow$ function space

0.1 $\rightarrow$

weak prediction

$$m_\eta = m_0 - \eta \text{ slope}$$

$\rightarrow$

$$f_m(x) = f_{m-1}(x) + \eta h_m(x)$$

$\rightarrow$ decision

learning rate